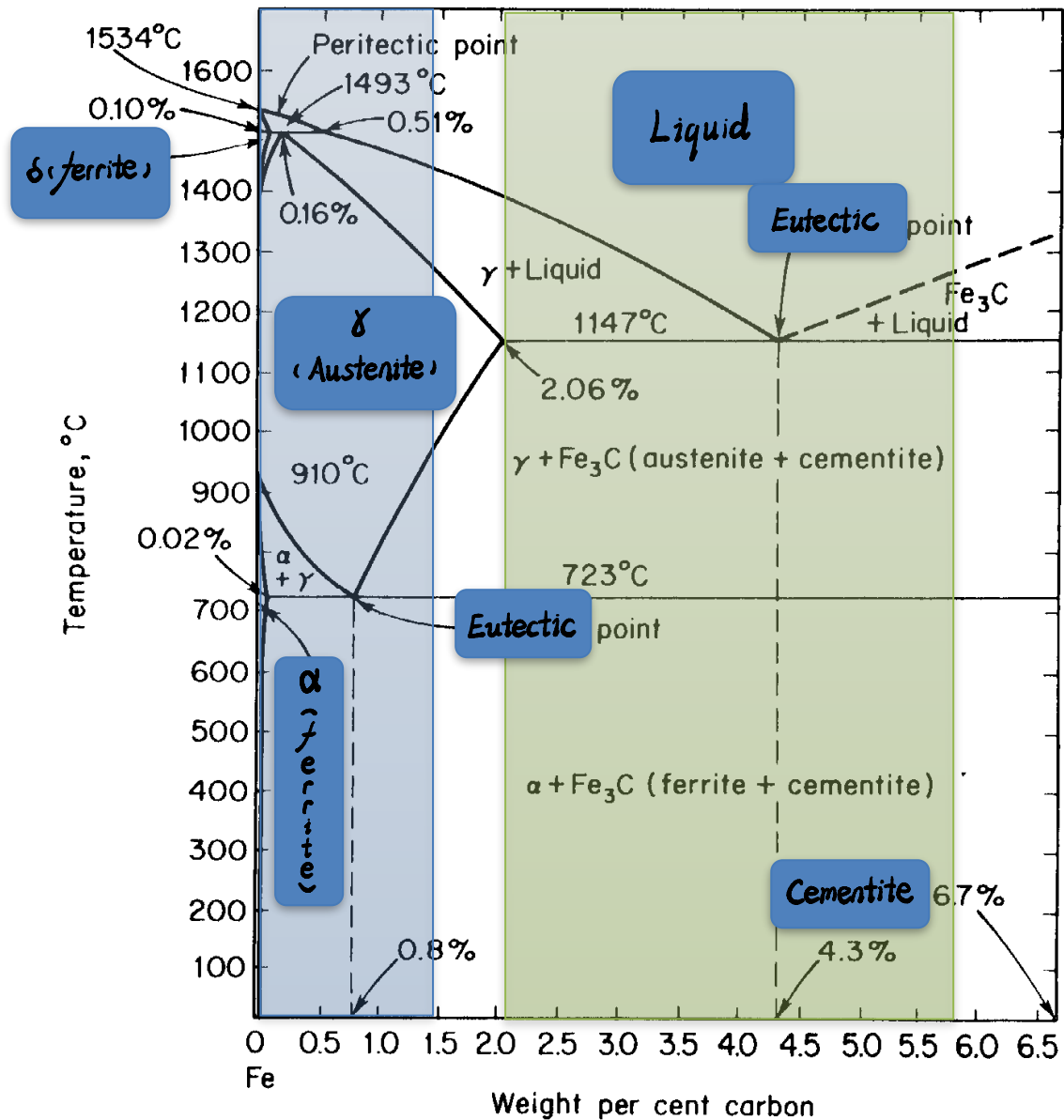


2. Alloys: Steel

1. Iron-carbon phase diagram

Complete the Iron-Carbon phase diagram below:



The iron alloys associated with the blue area are called carbon steels and with the green area are called cast irons.

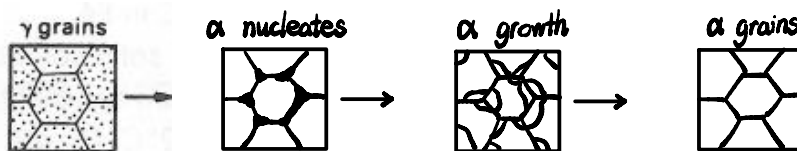
Important phases in steels:

Phase	Atomic packing	Description and comments
Liquid δ (ferrite)	d.r.p. b.c.c.	Liquid solution of C in Fe. Random interstitial solid solution of C in b.c.c. Fe. Maximum solubility of 0.08 wt% C occurs at 1492°C. Pure δ Fe is the stable polymorph between 1391°C and 1536°C (see Fig. 2.1).
γ (also called austenite)	f.c.c.	Random interstitial solid solution of C in f.c.c. Fe. Maximum solubility of 1.7 wt% C occurs at 1130°C. Pure γ Fe is the stable polymorph between 914°C and 1391°C (see Fig. 2.1).
α (also called ferrite)	b.c.c.	Random interstitial solid solution of C in b.c.c. Fe. Maximum solubility of 0.035 wt% C occurs at 723°C. Pure α Fe is the stable polymorph below 914°C (see Fig. 2.1).
Fe_3C (also called "iron carbide" or "cementite")	Complex	A hard and brittle chemical compound of Fe and C containing 25 atomic % (6.7 wt%) C.

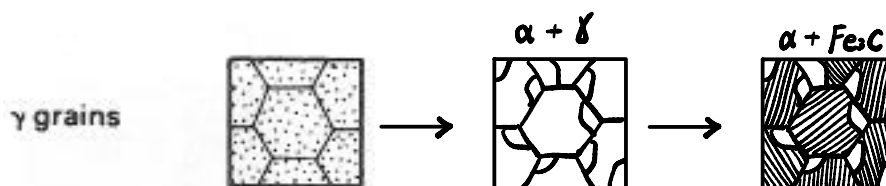
2. Diffusive phase transformation

Sketch the diffusive transformation when carbon steel is cooled from 1000° C down to room temperature, for:

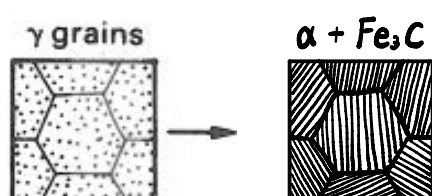
0 wt% carbon content (pure iron),



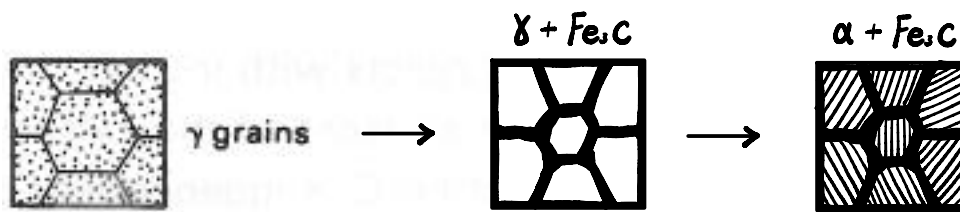
0.5 wt% carbon content (hypo-eutectoid steel),



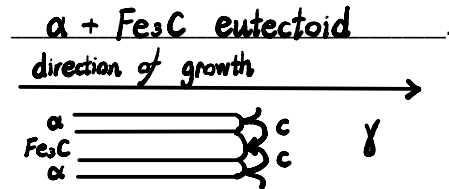
0.8 wt% carbon content (eutectoid steel),



1.0 wt% carbon content (hyper-eutectoid steel),

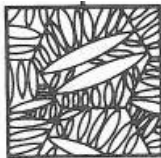


Define "pearlite" in the above schematic drawing.

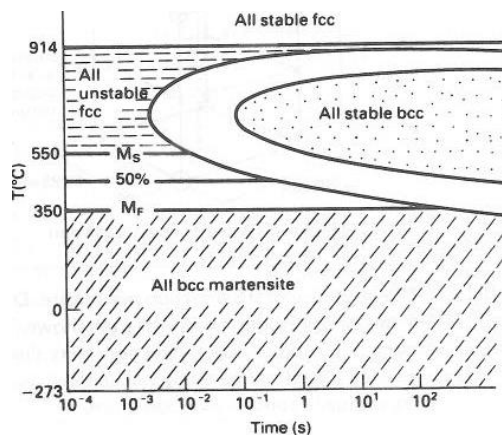


3. Displacive transformation (Martensitic)

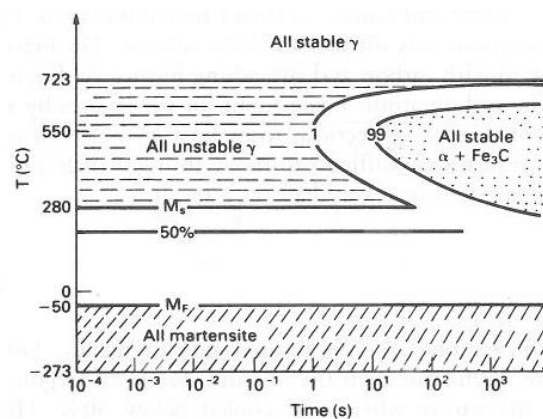
Displacive (martensitic) transformation occurs when cooling from the γ -FCC-Fe field fast enough, in which strained lenses of BCC-Fe form and grow throughout grains.



When increasing the carbon content in steel, it promotes the displacive transformation. The cooling rate to achieve such transformation is slower compared to pure iron. The hardness of martensite increases when carbon content increases, due to more carbon distort BCC lattice. However, ductility is decreasing.

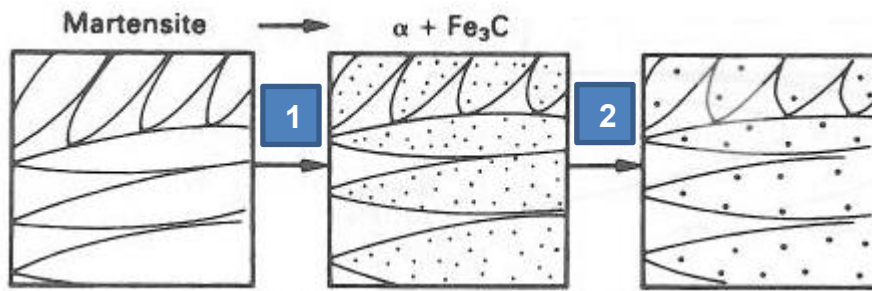


Pure iron



Fe-0.8wt%C

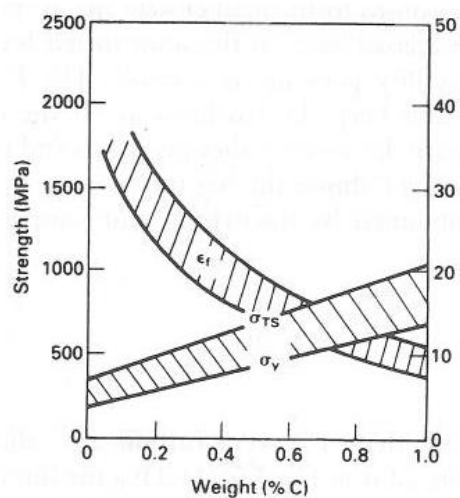
4. Tempering and coarsening



Process one is called tempering (reheat the steel to 300-600 °C). The black small dots are precipitates. By this heat treatment, the ductility of martensite can be improved because BCC lattice relax back to BCC lattice. Meanwhile, the strength/hardness can be maintained because of precipitation strengthening effect.

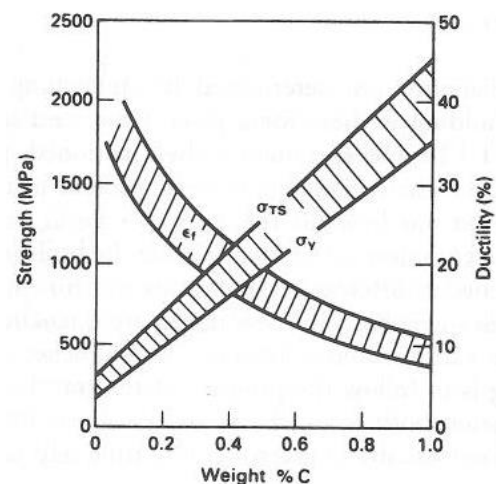
However, when we hold the steel at 300-600 °C for too long, Process two coarsening will happen, which will reduce the strength.

5. Mechanical properties of carbon steels



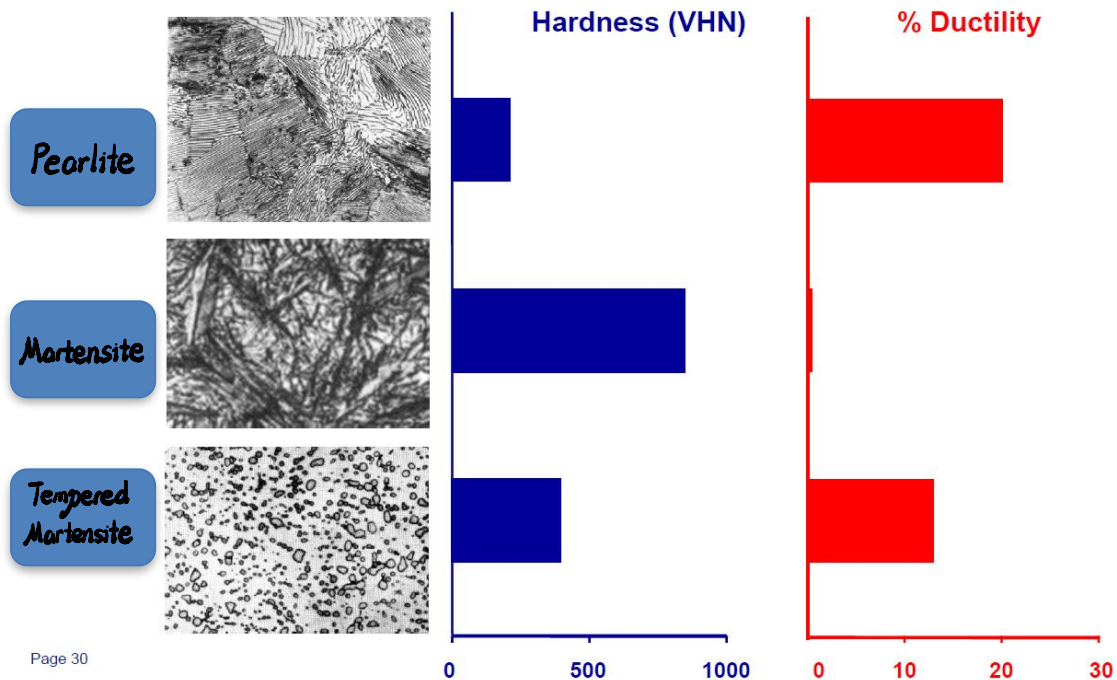
“Diffusive”:

process: slow cooled and normalised



“Displacive”

Quenched and tempered



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The fraction of pearlite, as well as Fe_3C increases/decreases with carbon content. The yield strength and UTS increase/decreases with carbon content because Fe_3C is a hard/soft phase. The ductility increase/decreases with carbon content because $\alpha\text{-Fe-Fe}_3\text{C}$ interfaces are good at nucleating cracks.

Martensite is very strong which gives very good hardness but low ductility. Tempered Martensite ($\alpha + \text{Fe}_3\text{C}$) offers a balanced strength and ductility.

Many engineering steels are 'medium carbon steels' 0.25-0.45 wt.%C quenched and tempered (displacive transformation), e.g. aircraft landing gears.

6. Alloy steels

For alloy steels, the alloying strategies are:

- To improve the hardenability of the steel, which is the ability of steel to form Martensite on quenching. By adding additional elements such as Mo, Mn, Cr and Ni, Martensite can be formed even at a lower cooling rate, so the surface and the interior of a component can both form martensite, even when air-cooled at room temperature.
- To give "solid solution strengthening" and "precipitation hardening", in which for the later effect, a number of elements are added to steels to form carbides, which are thermodynamically more stable than cementite and good for high temperature application.
- To give corrosion resistance such as adding Cr.
- To stabilise Austenite at room temperature (i.e. FCC steel) by adding elements such as Mn and Cr which are good α -ferrite stabiliser.